

Team-Based Negotiation

Negotiation has been studied widely, but it has been focused on processes whose parties (bilateral, or multiparty) are formed by a single individual.

However, some real world scenarios bring about negotiation parties that are formed by more than a single individual. For instance, when an organization negotiates with another organization the selling of a product line, it is usual for organizations to send a group of representatives to negotiate with the other organization. Another example, probably a more quotidian example, involves a married couple that negotiates the purchase of a house with a seller.

This kind of multi-individual party is known in the social sciences as a **negotiation team**: a group of interdependent people that join and act together as a single negotiation party because of their shared interests, related to a negotiation.

(Paragraphs extracted directly from [Anguix et al. \(2016\)](#))

Anguix et al.'s work has covered this idea, but I have not yet found studies that explore this setting in the context of LLM-based negotiation systems.

Pre-LLM precedents

[Anguix et al. \(2013\)](#) , [Anguix et al. \(2016\)](#)

NegotiationArena Scenarios

BuySell (incomplete-info), Trading (multi-issue, positive-sum) and Ultimatum (zero-sum split)

The scenarios in NegotiationArena aren't complex, so the team-negotiation should also take that into account, i shouldn't overcomplicate it!

Possible Dimensions

- Team Size

1 (baseline), 2, 3

- Model Heterogeneity

Same model for all vs mixed

- Role Structure

How tasks are divided across team members.

1. Symmetric/no-role
2. Leader + Analysts,

Inspiration

- Leader: Owns the strategy; is the designated speaker
- Analyst: Watches numbers, terms and trade-offs; tracks concessions

3. Leader + Analyst + Devil's Advocate

- a. Devil's Advocate prompted to challenge the team's draft (idea from

DEBATE: Devil's Advocate-Based assessment and text evaluation(Kim et al., ACL Findings 2024))

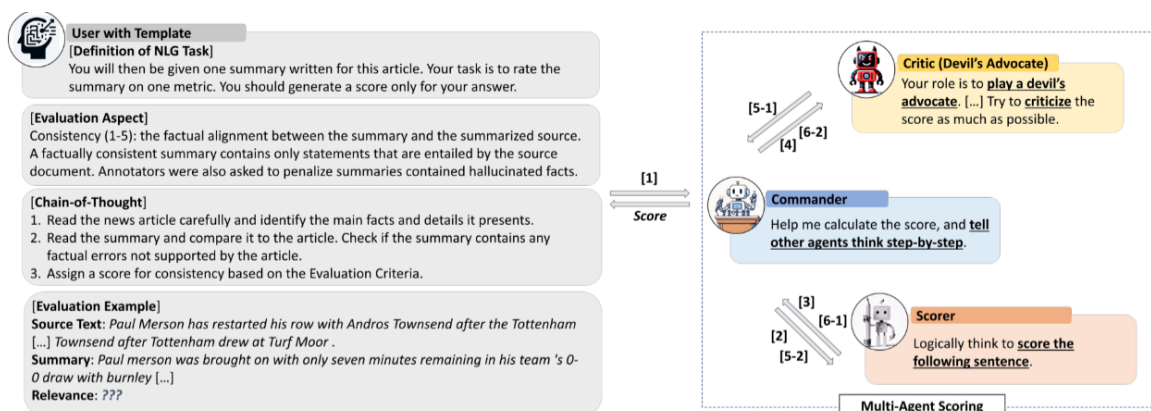


Figure 1: Overall framework of DEBATE. Numbers around the arrows correspond to the steps in Appendix D.1. The figure illustrates an example of deriving a consistency score in summarization task.

4. Good-cop/Bad-cop

The Good Cop, Bad Cop Negotiation Strategy: Two individuals, working as a team, extend a series of rewards and punishments with the goal of gaining an advantage over their counterpart.

This one looks interesting, but the classic negotiation involves the two agents speaking with the other party (we don't necessarily have consensus). I'm afraid that this would make it confusing for the other side of the negotiation.

A possible solution for that is still having a single agent as a speaker but to follow either of this:

3. **One person as both good cop and bad cop:** A solo negotiator uses a mixture of good cop and bad cop techniques, such as switching from friendly to impatient or being indecisive about what to offer.
4. **Good cop warning about a future bad cop:** A negotiator might warn that he's giving you the best deal possible and that if you come back tomorrow, you should expect to get a worse deal from his boss

- **Consensus**

How the team aggregates its members' positions into a single action.

1. Representative (RE) (from [Anguix et al. \(2016\)](#))
The leader hears all caucus contributions and decides unilaterally.
2. Similarity Simple Voting (SSV) (from [Anguix et al. \(2016\)](#))
Each team member independently votes on offers via a trusted mediator, with the winning option selected by plurality (for proposals) or majority (for acceptance).
3. Similarity Borda Voting (SBV) (from [Anguix et al. \(2016\)](#))
Each member ranks all candidate moves; Borda count aggregates.
4. Unanimity / Full Unanimity Mediated (FUM) (from [Anguix et al. \(2016\)](#))
Iterated, attribute-by-attribute negotiation until all members agree on every attribute

Intra-Team Strategy	Unanimity level (How)	Pre-negotiation? (When)	Mediated?
RE	Unilateral	Minimum	No
SSV	Plurality/Majority	Minimum	Only coordination
SBV	Semi-Unanimity	Minimum	Only coordination
FUM	Unanimity	Information sharing	Very Active

Table 1: A brief qualitative comparison among the four intra-team strategies presented in this article

5. Confidence-weighted vote (used on [RECONCILE \(Chen et al. 2024\)](#))
Each member submits a proposal and a self-reported confidence; weighted aggregation. LLM-specific extension that exploits calibration signal humans cannot give.

6. LLM-judge

A separate judge model reads all member proposals and synthesizes the team's external action. Differs from RE since the judge is not a team member and is meant to be neutral.

● Topology

1. Star/hierarchical (leader at hub)

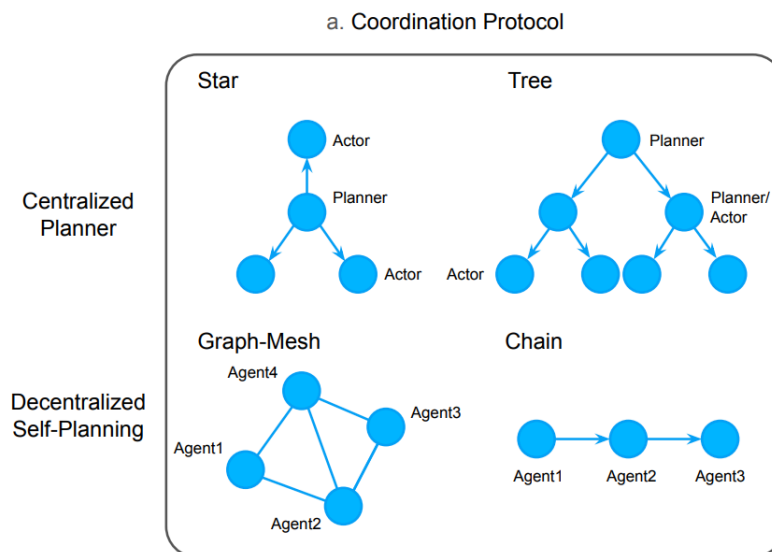
Only the Leader reads proposals; only the Leader communication with the counterpart

2. Mesh/egalitarian (all-to-all)

Each member sees each other's contribution before producing its own (iterative) or in parallel (single-shot).

3. Chain

Member 1 produces output, member 2 reads it and produces output, member 3 reads both. Models a review chain.



[Zhu et al. 2025](#)

● Personas

Behavioral or motivational priors injected into the prompt.

1. Team-level (single persona for the side)

- (the cunning/desperate personas in [Bianchi et al. \(2024\)/NegotiationArena](#))

2. Individual member-level (each member with its own OCEAN profile / risk tolerance)
 - a. [Noh et al. \(2024\)](#) analyzed how Big-Five OCEAN traits impact fairness-vs-rationality in single-agent negotiation.

Observations from some other papers to have in mind

- **Expertise dilution:** [Pappu et al. \(2024\)](#) noted that “LLM teams consistently fail to match their expert agent’s performance, even when explicitly told who the expert is, incurring performance losses of up to 37.6%.”. LLMs have a tendency toward compromise (averaging expert and non-expert views rather than appropriately weighting expertise)